HARD ALGEBRA

Linear functions

02

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ALGEBRA

x	-11	-10	-9	-8
f(x)	21	18	15	12

The table above shows some values of x and their corresponding values $f(x)$ for the linear function f. What is the x-intercept of the graph of $y=f(x)$ in the xy-plane?

A. $(-3,0)$

B. $(-4,0)$

C. $(-9,0)$

D. $(-12,0)$

Q2 ALGEBRA

A window repair specialist charges $\$ 220$ for the first two hours of repair plus an hourly fee for each additional hour. The total cost for 5 hours of repair is $\$ 400$. Which function f gives the total cost, in dollars, for x hours of repair, where $x \geq 2$?

A. $f(x) = 60x + 100$

B. $f(x) = 60x + 220$

C. $f(x) = 80x$

D. $f(x) = 80x + 220$

Q3 ALGEBRA

One gallon of paint will cover 220 square feet of a surface. A room has a total wall area of w square feet. Which equation represents the total amount of paint P, in gallons, needed to paint the walls of the room twice?

A. $P = \frac{w}{110}$

B. $P = 440w$

C. $P = \frac{w}{220}$

D. $P = 220w$

Q4 ALGEBRA

The linear function g is defined by $gx = b$ - $15x$, where b is a constant. If $gc + 7 = \frac{c}{4}$, where c is a constant, which of the following expressions represents the value of b?

A. $\frac{15c}{4}$

B. $\frac{19c}{4} + 7$

C. $\frac{61c}{4} + 105$

D. $15c + 105$

Q5 ALGEBRA

The cost of renting a large canopy tent for up to 10 days is $\$ 430$ for the first day and $\$ 215$ for each additional day. Which of the following equations gives the cost y, in dollars, of renting the tent for x days, where x is a positive integer and $x \leq 10$?

A. $y = 215x + 215$

B. $y = 430x$ - 215

C. $y = 430x + 215$

D. $y = 215x + 430$

Q6

ALGEBRA

According to data provided by the US Department of Energy, the average price per gallon of regular gasoline in the United States from September 1, 2014, to December 1, 2014, is modeled by the function F defined below, where

$$F(x)$$

is the average price per gallon x months after September 1.

$$F(x) = 2.74 - 0.19(x - 3)$$

The constant 2.74 in this function estimates which of the following?

A. The average monthly decrease in the price per gallon

B. The difference in the average price per gallon from September 1, 2014, to December 1, 2014

C. The average price per gallon on September 1, 2014

D. The average price per gallon on December 1, 2014

Q7

ALGEBRA

$$Fx = \frac{9}{5}x - 273.15 + 32$$

The function F gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. If a temperature increased by 9.10 kelvins, by how much did the temperature increase, in degrees Fahrenheit?

A. 16.38

B. 48.38

C. 475.29

D. 507.29

Q8 GRID-IN ALGEBRA

The functions f and g are defined as $f\left(x\right) = \frac{1}{4}x - 9$ and $g\left(x\right) = \frac{3}{4}x + 21$. If the function h is defined as $h(x) = f(x) + g(x)$, what is the x-coordinate of the x-intercept of the graph of $y = hx$ in the xy-plane?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

ALGEBRA

For groups of 25 or more people, a museum charges $\$ 21$ per person for the first 25 people and $\$ 14$ for each additional person. Which function f gives the total charge, in dollars, for a tour group with n people, where $n \geq 25$?

A. $f n = 14n + 175$

B. $f n = 14n + 525$

C. $f n = 35n$ - 350

D. $f n = 14n + 21$

Q10 ALGEBRA

The cost of renting a backhoe for up to 10 days is \$ 270 for the first day and \$ 135 for each additional day. Which of the following equations gives the cost y, in dollars, of renting the backhoe for x days, where x is a positive integer and $x \leq 10$?

A. $y = 270x - 135$

B. $y = 270x + 135$

C. $y = 135x + 270$

D. $y = 135x + 135$